

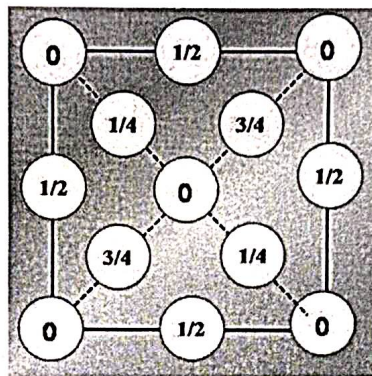
End-Term (PART-A)

COURSE: Characterization of Materials

Total time: 60 mins

Answer All the Questions (Total = 20)

1. The below image is the AI predicted 2D Unit Cell Projection of an unknown element. Retrieve required crystallographic data from this image and show the first four possible reflection (diffraction) peak for an XRD Characterization Experiment for this unknown element. Now show the mathematical scheme for your answer in details. (3+7=10)



2. The Below simulated diffraction pattern has been formulated from experimental information of a binary alloy system with A and B. Now index the planes which are not assigned with any Miller Indices. Find out the Zone Axis. Draw the Probable Unit Cell of the structure and establish the orientation relationship with the newly indexed planes with respect to the already indexed planes. Derive the (hkl) extinction rule of diffraction for this unknown structure. (1+1+3+5=10)

